AI-Supported Development of Model Exam Paper and Aligned Short Notes

Project Id -25-26J-089

Project Proposal Report

Leashaniya Krishnapillai - IT22891440

Supervisor – Dr. Sanvitha Kasthuriarachchi

Co-Supervisor - Ms. Karthiga Rajendran

BSc (Hons) in Information Technology Specialized in Data Science

Department of Information Technology

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DECLARATION OF THE CANDIDATE & SUPERVISOR

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Leashaniya K	IT22891440	

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

--- ---

Signature of the supervisor
(Dr. Sanvitha Kasthuriarachchi)

Date

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Signature of co-supervisor
(Ms. Karthiga Rajendran)

Date

Abstract

Universities frequently face challenges in aligning lecture delivery with examination patterns, often causing students to revise less relevant topics while overlooking frequently tested areas. Manual reviews of past exam papers and lecture notes are time-consuming, inconsistent, and subjective, highlighting the need for a systematic, data-driven approach.

This research proposes an Agentic AI framework to automate the analysis of historical exam papers and lecture materials in modules. The framework consists of specialized agents: an Ingestor Agent for preprocessing, an Exam Analyst Agent for detecting exam structures and mark distributions, and a Trend Miner Agent applying NLP techniques such as TF-IDF, key phrase extraction, and semantic clustering to uncover high-priority topics across 8–10 years of assessments. Using these insights, the Generator Agent produces a synthetic model exam paper that authentically mirrors historical formats, question styles, and difficulty levels. The Mapper Agent then links model paper topics to lecture materials, enabling the Notes Agent to generate concise short notes directly aligned with the topics in the model paper. An Evaluator Agent validates duplication, coverage, difficulty balance, and alignment with academic standards.

The novelty of this component lies in the automated generation of concise, curriculum-aligned short notes that are strictly mapped to a synthetically generated model exam paper. Unlike conventional broad-scope revision materials, the proposed framework delivers narrowly focused, evidence-based resources that reflect actual historical exam trends, thereby enhancing exam preparation efficiency and curriculum alignment.

Keywords: Agentic AI, exam structure analysis, topic frequency mining, NLP, model exam paper generation, curriculum-aligned short notes

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List of Abbreviations

Abbreviations	Description
ML	Machine learning
NLP	Natural language processing
BERT	Bidirectional Encoder Representations from Transformers
SBERT	Sentence Bidirectional Encoder Representations from Transformers
LLM	Large Language Model
TF-IDF	Term Frequency–Inverse Document Frequency
API	Application Programming Interface
TEL	Technology Enhanced Learning

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01. Introduction

Assessment is one of the most influential drivers of student learning in higher education, yet many universities struggle to maintain consistent alignment between curriculum delivery, lecture content, and examination practices. This misalignment often results in students devoting significant time to topics that appear only rarely in exams, while overlooking those that are frequently tested and more critical for success. Therefore, students' preparation becomes inefficient and learning outcomes are not fully achieved. Traditional approaches, such as manually analyzing past exam papers and lecture materials, are often subjective, inconsistent, and time-consuming, making them inadequate for addressing this issue on a larger scale.

The emergence of Artificial Intelligence (AI), Natural Language Processing (NLP), and educational analytics presents a timely opportunity to transform this process. Unlike conventional learning management systems, which largely focus on content delivery and administrative functions, AI-powered approaches can uncover structural and topical trends across years of academic assessments, identify gaps between taught material and examined content, and provide actionable insights for both learners and educators. These capabilities not only optimize revision strategies for students but also guide institutions in improving curriculum design and instructional effectiveness.

Our research targets the development of an intelligent, AI-powered academic support framework that systematically analyzes historical exam papers, lecture content, and continuous assessments to deliver data-driven insights and personalized learning outputs. Through advanced NLP techniques, the system identifies high-frequency topics, predicts question difficulty, and classifies content according to syllabus requirements, while also highlighting redundancies and underrepresented areas. Based on these analyses, the framework generates practical learning aids, including model exam papers that reflect historical patterns, concise and curriculum-aligned short notes, adaptive essay evaluation with feedback, interactive flashcards, voice notes, and personalized time-aware study schedules.

By integrating these elements into a unified platform, the research provides a holistic solution that enhances student self-learning, improves exam preparation efficiency, and strengthens curriculum alignment with assessment practices. Ultimately, the project seeks to bridge the gap between academic delivery and evaluation through a data-driven, scalable, and personalized educational support system, thereby contributing to improved student achievement and institutional teaching quality.

1.1 Background and Literature Review

1.1.1. Background

Assessment is one of the most influential drivers of student learning in higher education. Examination papers not only evaluate students but also act as implicit indicators of curricular priorities. However, several studies have noted frequent misalignment between lecture delivery and exam content, leading students to inefficiently allocate study time by overemphasizing rarely tested topics while underpreparing for frequently recurring areas. This ultimately affects academic performance and weakens curriculum effectiveness [1].

Traditional exam preparation and curriculum review methods depend heavily on manual inspection of past papers and lecture notes. While useful at a small scale, such approaches are time-consuming, subjective, and inconsistent across evaluators [2]. With the rapid expansion of digital learning content, these manual approaches are no longer sustainable or scalable. This has prompted growing interest in data-driven educational analytics that systematically analyze academic assessments and instructional materials to guide evidence-based curriculum alignment and student revision strategies [3].

In this context, Artificial Intelligence (AI) and Natural Language Processing (NLP) have become powerful enablers. Techniques such as keyword extraction, topic frequency analysis, semantic clustering, and text summarization facilitate efficient exploration of large collections of exam questions and lecture documents [4]. Transformer-based models, including BERT, T5, and Sentence-BERT, enable semantic similarity analysis between lecture content and exam questions, making it possible to align curriculum delivery with assessment patterns more accurately [5]. These advancements create opportunities for designing intelligent systems that integrate exam trend mining with curriculum mapping**,** ultimately producing targeted study resources such as model exam papers and syllabus-aligned short notes.

1.1.2. Literature Review

Research on Automatic Question Generation (AQG) has long been explored in educational NLP. Early systems were rule-based and template-driven, which ensured grammatical correctness but lacked scalability and semantic depth [6]. Neural and transformer-based methods improved contextual relevance, but most AQG studies focused on linguistic quality rather than enforcing exam-specific structures such as section divisions, mark distributions, or repeated patterns. As a result, these methods fall short of generating authentic model exam papers that reflect institutional assessment blueprints [7].

Beyond AQG, work on short answer grading and feedback generation has advanced from keyword matching to semantic embedding, allowing deeper analysis of student responses [8]. Technology Enhanced Learning (TEL) frameworks have even proposed integrated pipelines combining question generation, grading, and adaptive feedback. However, these systems generally overlook long-term exam trend analysis and rarely connect generated outputs back to curricular content [9].

Parallel research in summarization and topic modeling aims to condense large volumes of instructional content into focused study resources. Extractive techniques such as TF-IDF and TextRank ensure factual coverage of subtopics, while abstractive approaches such as BART and T5 produce more natural summaries [10]. Recent improvements in contextual embeddings (e.g., Sentence-BERT) enable semantic matching between topics and lecture material [11]. Nevertheless, most approaches lack mechanisms to prioritize content based on exam frequency trends, leading to generic summaries rather than exam-oriented short notes.

With the rise of Large Language Models (LLMs), controllable and rubric-aware generation has attracted interest. Prompts and control tokens can influence difficulty levels and scope, and LLM evaluators are used for automated scoring [12]. Yet these systems often face reproducibility issues, limited explainability, and weak alignment with institutional rubrics, making them unsuitable for generating reliable model exam resources [13].

In summary, while prior research provides strong foundations in question generation, summarization, and adaptive feedback, significant gaps remain in combining:

1. Exam structure fidelity (sections, marks, formats).
2. Long-term exam trend mining to identify high-priority topics.
3. Semantic mapping between model exam questions and lecture materials.
4. Automated generation of curriculum-aligned short notes strictly linked to model papers.

Addressing these gaps, this research introduces an Agentic AI-driven approach that integrates exam trend analysis, semantic mapping, and summarization into a unified pipeline to produce realistic model exam papers and targeted short notes aligned with curricular needs.

1.2 Research Gap

Although there has been substantial progress in automatic question generation (AQG), summarization, and educational analytics, existing research still falls short in several key areas relevant to structured university examinations. Most prior studies on AQG focus on generating linguistically correct questions but rarely incorporate exam-specific structures, such as section divisions, mark distributions, or recurring question formats [14], [16]. Similarly, educational summarization systems condense lecture materials but fail to prioritize content based on exam frequency trends, resulting in generic summaries that are not directly aligned with historical patterns [16].

Furthermore, while Large Language Models (LLMs) have introduced controllable question generation and automated evaluation, they often lack reproducibility, rubric alignment, and institution-specific customization, making them unsuitable for producing authentic model exam papers [17]. Answer evaluation pipelines support formative assessment but are limited in their ability to validate exam-quality questions or forecast emerging topics [15].

This gap indicates the absence of a system that can:

- Conduct longitudinal analysis of past papers to detect structural and topical patterns.
- Generate synthetic model exam papers that authentically mirror historical formats and difficulty levels.
- Produce short notes strictly mapped to model exam paper topics, ensuring curriculum alignment and targeted revision.
- Forecast future exam themes using statistical and NLP-driven trend analysis.

Addressing these challenges, the proposed component introduces an Agentic AI-driven framework that uniquely integrates exam structure analysis, statistical topic mining, semantic lecture alignment, and automated summarization. The novelty lies in producing concise short notes directly aligned to synthetically generated model exam papers, providing a reliable, evidence-based, and exam-oriented resource for students.

Research Gap Comparison

Table 1 – Research gap comparison

Feature / Capability	**Research [14]**	**Research [15]**	**Research [16]**	**Research [17]**	**Proposed System**
Question Generation	✓ Generic	✓ With answer grading	✓ Controllable prompts (difficulty only)	✓ LLM-based generation	✓ Exam-specific model exam papers
Exam Structure Fidelity	✗	✗	✗	✗	✓ Explicit structure analysis + blueprint alignment
Summarization	✗	✗	✓ Generic summarization	Partial	✓ Concise short notes mapped to syllabus
Lecture Alignment	✗	✗	✗	✗	✓ Direct mapping of notes to model paper topics
Topic Trend Forecasting (multi-year)	✗	✗	✗	✗	✓ Statistical + NLP-driven trend analysis
Automated Short Notes Generation	✗	✗	✗	✗	✓ Automated, curriculum-aligned short notes

1.3 Research Problem

Assessment remains one of the most influential drivers of student learning in higher education. Examination papers not only serve as instruments of evaluation but also implicitly signal curricular priorities. However, numerous studies highlight a persistent misalignment between lecture content delivery and examination focus, which leads students to inefficiently allocate study time toward topics with low exam frequency while neglecting high-priority areas [1], [2]. Traditional approaches to bridging this gap have relied on manual analysis of past papers and lecture materials, which are time-consuming, subjective, and inconsistent across reviewers [3].

Recent developments in automatic question generation (AQG), educational summarization, and learning analytics demonstrate strong potential for supporting personalized learning. Yet, as identified in prior surveys, these methods remain limited in several critical ways:

1. **AQG systems** focus on linguistic quality but rarely enforce exam-specific structures such as section divisions, mark allocations, and recurring formats [14], [16].
2. **Automated evaluation systems** provide answer grading but lack curriculum-aligned validation, making them unsuitable for high-stakes assessments [15].
3. **Summarization models** condense lecture materials effectively, but they do not prioritize topics based on historical exam frequency, resulting in generic notes rather than exam-focused resources [16].
4. **LLM-based question generation** offers controllability, but current methods suffer from low reproducibility, weak rubric alignment, and limited institutional customization in educational settings [17].

Consequently, there exists a clear gap in developing an integrative system that not only automates past-paper analysis but also forecasts topic trends, generates synthetic model exam papers that reflect authentic historical structures, and produces concise short notes directly mapped to high-frequency syllabus content. Addressing this research problem is vital for optimizing exam preparation, ensuring stronger alignment between teaching and assessment, and providing students with reliable, personalized, and data-driven revision resources.

02. Objectives

2.1 Main objective

To design and implement an **Agentic AI-powered framework** that analyzes historical exam papers to identify structural and topical patterns, generates synthetic model exam papers, and produces curriculum-aligned short notes directly mapped to lecture materials, thereby enhancing student revision efficiency.

2.2 Specific objective

- Data Collection & Preprocessing (Ingestor Agent): Collect and preprocess 8–10 years of past exam papers and lecture materials using OCR, cleaning, and tokenization to build a structured corpus.
- Exam Structure Analysis (Exam Analyst Agent): Automatically detect section patterns, mark distributions, and question formats using rule-based and NLP techniques.
- Topic Frequency & Trend Mining (Trend Miner Agent): Apply NLP techniques such as TF-IDF, keyphrase extraction, and semantic clustering to identify high-priority and recurring exam topics.
- Model Exam Paper Generation (Generator Agent): Generate synthetic model exam papers that realistically mirror historical exam structures, difficulty levels, and topic distributions.
- Semantic Mapping to Lecture Content (Mapper Agent): Link topics identified in the model exam paper with corresponding lecture materials using embedding-based similarity search.
- Short Notes Generation (Notes Agent): Produce concise, syllabus-aligned short notes from mapped lecture content using a combination of extractive and abstractive summarization.
- Evaluation & Validation (Evaluator Agent): Validate system outputs by checking duplication, coverage, difficulty balance, and alignment with academic standards.

03. Methodology

3.1 Requirement Gathering

The requirement gathering process for this project involved analyzing existing research, identifying the limitations of current educational support systems, and understanding the specific needs of university students preparing for exams. This phase focused on identifying data requirements, functional expectations, and technological considerations necessary for developing an AI-driven solution capable of generating model exam papers and curriculum-aligned short notes.

3.1.1 Past Research Analysis

A detailed review of past studies on exam paper analysis, question generation, and educational summarization was carried out to understand the strengths and weaknesses of existing methods. Prior research revealed that while NLP techniques can automate question generation and summarization, they often lack exam-structure fidelity and syllabus alignment. The findings highlighted the need for an integrated framework capable of combining exam trend analysis, structural blueprinting, and lecture mapping to produce exam-oriented resources.

3.1.2 Identification of Existing Systems

An evaluation of current academic support platforms (such as generic summarization tools and e-learning systems) was performed to identify their limitations. Most platforms provide broad summaries or question banks but fail to capture authentic exam structures or prioritize high-frequency topics. This study confirmed the necessity of a system that goes beyond surface-level content generation by ensuring alignment with historical exam trends and academic curricula.

3.2 Development Methodology

To effectively manage the development of the proposed Agentic AI framework for model exam paper generation and curriculum-aligned short notes, factors such as time, scope, and flexibility were carefully considered. The project must handle evolving requirements, especially as models are tested and refined to ensure exam-structure fidelity and syllabus alignment. After evaluating different approaches, the Agile methodology was selected to guide the development process.

This methodology supports an iterative and incremental development cycle, where each sprint focuses on a specific agent in the framework (e.g., Ingestor Agent for preprocessing, Trend Miner Agent for topic analysis, Generator Agent for model paper creation, Notes Agent for short notes). By developing and testing these modules in short iterations, early prototypes can be demonstrated to the supervisor for feedback.

Regular feedback during sprints ensures that issues such as topic misalignment, summarization accuracy, or exam blueprint fidelity are identified and corrected early. This reduces rework and improves efficiency. The iterative cycle also allows seamless integration of new requirements or improvements suggested by the mentor, ultimately leading to a robust, validated system.

The Agile methodology therefore provides the flexibility to refine the system continuously while ensuring steady progress toward the final product: a functional prototype capable of producing synthetic model exam papers and concise, syllabus-aligned short notes that directly support efficient exam preparation.

3.3 Project management Methodology

Research projects require effective project management to ensure that scope, time, and cost remain aligned throughout the development process. For the proposed Agentic AI framework focusing on model exam paper generation and curriculum-aligned short notes, a process-oriented project management methodology will be adopted. This methodology divides the project into distinct phases, where each phase delivers specific outputs such as data preprocessing, exam structure analysis, topic trend mining, model paper generation, and short notes creation.

The process-oriented approach provides a structured pathway to manage the project effectively, while also allowing flexibility to accommodate changes in requirements during development. It ensures that risks—such as delays in data collection, model inaccuracies, or scope creep—are identified early and addressed promptly.

To support this methodology, JIRA will be used as the primary project management tool. JIRA enables efficient task allocation, sprint planning, and progress tracking across all development phases. Each agent in the system (e.g., Ingestor, Exam Analyst, Trend Miner, Generator, Mapper, Notes, Evaluator) will be managed as a modular task, making it easier to monitor progress, incorporate feedback, and integrate components seamlessly.

By leveraging JIRA and a structured project management methodology, the research will maintain clear accountability, transparent progress tracking, and alignment with project timelines, ultimately ensuring the successful delivery of a validated prototype.

3.4 Feasibility study

3.4.1 Technical Feasibility

To develop the proposed solution, several technological capabilities are required. The system is designed as an Agentic AI framework, where each specialized agent carries out a distinct task in the pipeline: preprocessing, exam structure analysis, trend mining, model paper generation, semantic mapping, short note generation, and output evaluation.+

Knowledge On Technologies:

- Natural Language Processing (NLP): Techniques for keyword extraction, topic frequency analysis, and semantic clustering to identify high-priority exam topics.
- Machine Learning Models: Familiarity with transformer-based models (e.g., BERT, T5, SBERT) for summarization, paraphrasing, and content alignment.
- Semantic Mapping: Use of embedding-based similarity (e.g., cosine similarity with Sentence-BERT + FAISS) to link exam topics with lecture materials.
- Summarization: Extractive (TextRank, TF-IDF) and abstractive (BART, PEGASUS, T5) methods for generating concise, curriculum-aligned short notes.
- Data Analysis Tools: Proficiency in handling large datasets using Python libraries such as Pandas and NumPy to analyze trends and frequencies.

Knowledge On Tools:

- Programming Languages: Python (NLP/ML pipeline), JavaScript (frontend).

- Frameworks: Flask/Django (backend), ReactJS (frontend).
- NLP & ML Libraries: Hugging Face Transformers, SpaCy, NLTK, scikit-learn.
- Vector Databases: FAISS for fast semantic similarity search.
- Data Analysis Libraries: Pandas, NumPy.
- Diagramming Tools: Draw.io, Lucidchart for architecture diagrams.
- Project Management Tools: JIRA for sprint planning, task tracking, and progress monitoring.

Data Collection Knowledge:

- Data Collection: Historical exam papers (8–10 years) and official lecture materials are identified as the core datasets.
- Data Preprocessing (Ingestor Agent): Includes OCR (for scanned documents), cleaning, tokenization, stopword removal, normalization, and error correction to prepare the data for downstream analysis and modeling.

By leveraging open-source tools and a modular Agentic AI architecture, the project is technically feasible, scalable, and adaptable to future requirements.

3.4.2 Schedule Feasibility

The project will follow a systematic timeline aligned with research milestones. Work will be divided into phases including data collection, preprocessing, model design, implementation, testing, and validation. Defined deliverables will ensure progress is tracked and deadlines are met without overruns.

3.4.3 Economic Feasibility

The system will be designed as a cost-effective solution by leveraging open-source NLP and machine learning libraries. Cloud-based or local deployment options will be used to minimize infrastructure costs. Budget considerations include computing resources, storage, and limited licensing for specialized tools if required. The approach ensures that the system can be developed and deployed with minimal financial overhead while maintaining scalability for future use.

3.5 System diagram

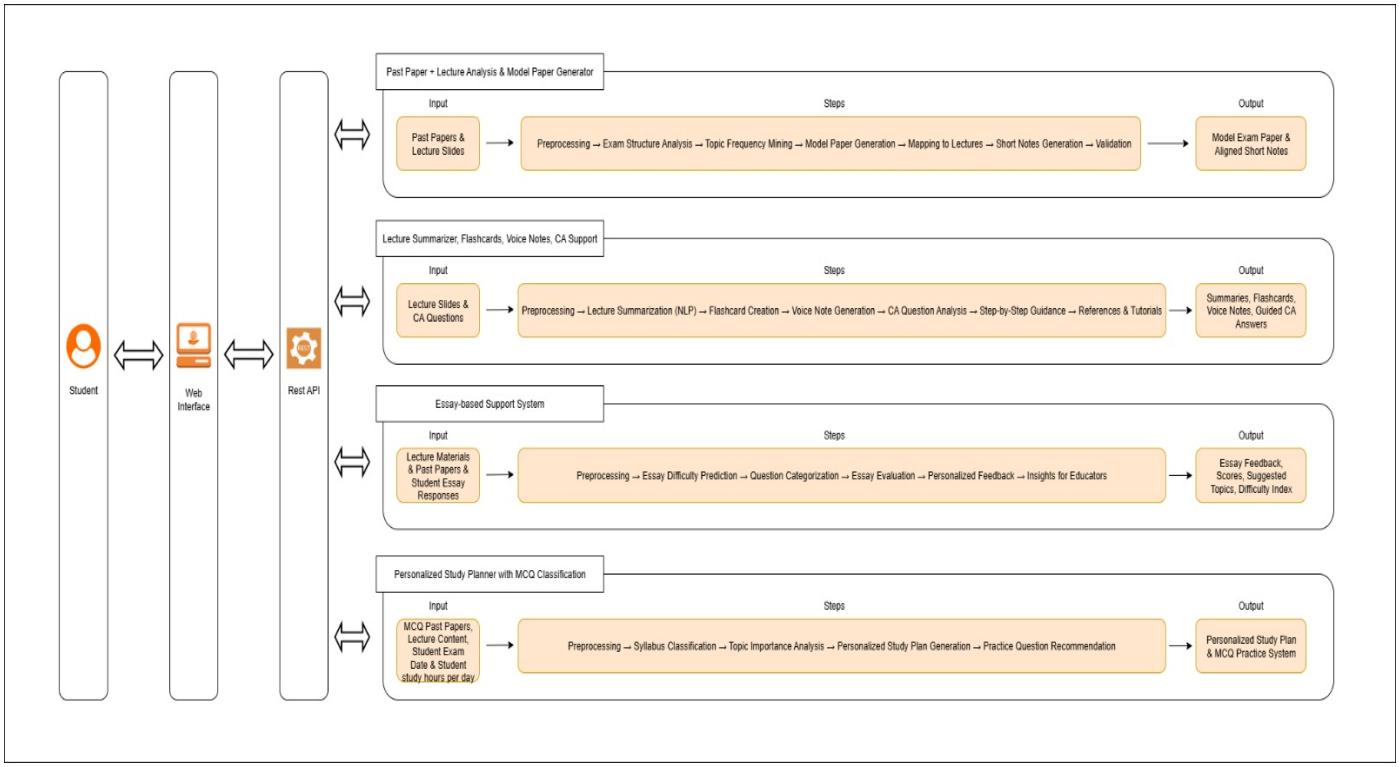

Figure 1 - System Diagram

3.6 System Analysis
3.6.1 Software Solution Approach

The development of the proposed system follows modular Agentic AI architecture, where each agent is responsible for a specific task in the pipeline. The approach involves the following steps:

- Data Collection & Preprocessing (Ingestor Agent): Collect 8–10 years of historical university exam papers and official lecture materials. Apply OCR (for scanned documents), text cleaning, tokenization, stopword removal, normalization, and error correction to prepare a structured dataset. Exam Structure Analysis (Exam Analyst Agent): Identify consistent patterns in exam papers such as section divisions, question formats, and mark allocations. This analysis provides the structural blueprint necessary for generating authentic model exam papers.
- Topic Extraction & Trend Mining (Trend Miner Agent): Apply NLP techniques such as keyword extraction, TF-IDF, and semantic clustering to identify high-frequency topics and recurring patterns across multiple years of exams. These insights prioritize topics for inclusion in model exam papers and short notes.
- Model Exam Paper Generation (Generator Agent): Construct synthetic model exam papers that authentically mirror historical structures, difficulty levels, and topic distributions. Paraphrasing models (e.g., FLAN-T5) are used to ensure novelty while maintaining semantic accuracy.
- Semantic Mapping to Lecture Content (Mapper Agent): Perform keyword-based and embedding-based similarity searches (e.g., Sentence-BERT + FAISS) to align identified exam topics with the most relevant lecture slides and notes.
- Short Notes Generation (Notes Agent): Automatically generate concise, curriculum-aligned short notes directly mapped to the model exam paper topics. A hybrid approach combines extractive summarization (TextRank, TF-IDF) with abstractive techniques (BART, PEGASUS) to ensure clarity and completeness.
- Validation & Evaluation (Evaluator Agent): Evaluate all outputs against exam blueprints and rubric standards. Checks include duplication avoidance, topic coverage, difficulty balance, and alignment with curriculum requirements, ensuring the reliability and academic relevance of the generated resources

3.6.2 Tools & Technologies

The following tools and technologies will be used for developing the system:

- Programming Languages: Python (core NLP/ML pipeline), JavaScript (frontend).
- NLP & Machine Learning Libraries: NLTK, SpaCy, Scikit-learn, Hugging Face Transformers (BERT, SBERT, T5, FLAN-T5, DistilBERT, BART, PEGASUS).
- Vector Search & Semantic Mapping: FAISS for efficient similarity search and topic–lecture alignment.
- Summarization & Topic Analysis: TextRank, TF-IDF for extractive summarization and frequency analysis.
- Data Analysis Tools: Pandas, NumPy, Matplotlib for data processing and visualization of topic trends.

- Web Development Frameworks: ReactJS (front-end), Flask/Django (back-end API).
- UI/UX Design Tools: Figma, Canva for interface prototyping and design.
- Diagramming Tools: Draw.io, Lucidchart for system and workflow diagrams.
- Integrated Development Environments (IDE): VS Code, PyCharm for coding and debugging.
- Version Control Systems (VCS): GitHub or GitLab for collaborative development and code management.
- Collaboration Tools: Microsoft Teams, Slack for communication and coordination.
- Testing Tools: PyTest, UnitTest, Postman (for API testing and validation).
- Deployment Tools: Docker (containerization), Kubernetes (orchestration), AWS/GCP (cloud hosting).
- Project Management Tools: JIRA (sprint planning, task tracking, and project monitoring).

3.7 Gantt chart

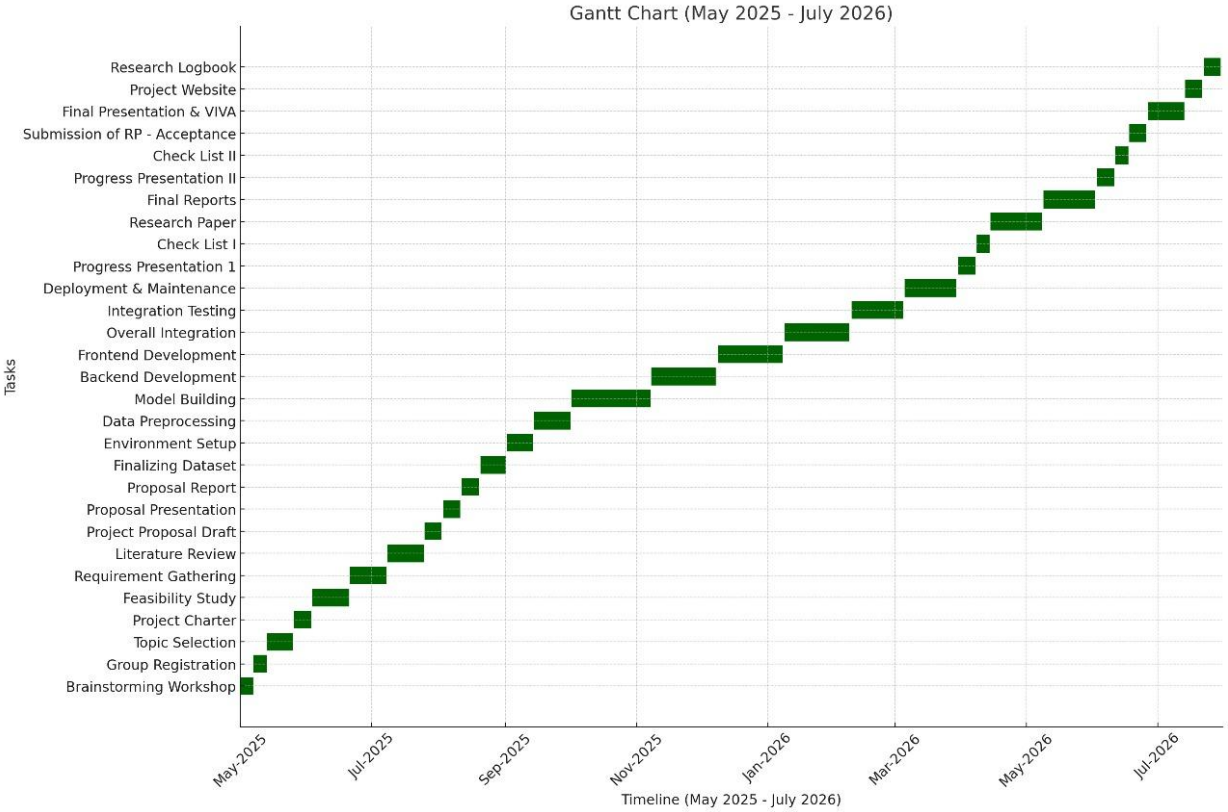

Figure 2 - Gantt Chart

4. Project Requirements

Functional Requirements

- Academic Data Collection & Preprocessing
  The system should collect past exam papers for the target module along with corresponding lecture notes. Documents should be preprocessed using text cleaning, and tokenization so they are ready for NLP analysis..

- Exam Structure Analysis
  The system should automatically detect and record exam patterns such as sections, mark distributions, and question types. This will be done using PDF parsing tools (like PyMuPDF or pdfplumber), rule-based text processing with regex, and NLP models to classify and segment questions.

- Topic Frequency and Trend Mining
  Natural Language Processing methods such as TF-IDF, keyword extraction, and semantic clustering should be used to identify frequently tested topics. Topics must be ranked according to frequency and weighted by mark allocation.

- Synthetic Model Exam Paper Generation
  A model exam paper should be generated that reflects historical exam structures and topic trends. It should include balanced topic coverage, realistic question styles, and appropriate mark distribution.

- Topic-to-Lecture Mapping
  The system should semantically match the topics in the model exam paper with relevant lecture materials.

- Short Notes Generation
  Concise, topic-focused short notes should be created directly from lecture materials. These notes should highlight key definitions, core ideas, and examples that are directly relevant to exam preparation.

- Output Validation and Review
  All generated model papers and short notes should go through a peer or mentor review process. A versioning mechanism must allow corrections and ensure quality before final outputs are approved.

Non Functional Requirements

- Performance
  The system should process exam papers and generate results within a reasonable time so users can quickly obtain model papers and notes without long delays.

- Reliability
  The system should operate consistently without frequent crashes or data loss.

- Scalability
  The framework should be able to manage multiple years of exam papers and lecture notes without reducing accuracy.

- Usability
  The interface should be simple and easy to use, presenting outputs such as model exam papers and short notes in a clear, structured format suitable for students and lecturers.

- Maintainability
  The system should have a modular design, making it easy to update or improve components such as preprocessing, topic mining, or short note generation without rebuilding the entire system.
- Security and Privacy
  Academic data should be stored securely and only accessed by authorized project members. Sensitive information must remain protected at all times.
- Educational Validity
  Generated content should align with the curriculum and learning outcomes. All outputs should be validated by academic mentors before being shared with students.

User requirements

Student (primary user)

- I need to upload or select my module's past papers and view the detected exam structure clearly.
- I need a model exam paper that mirrors real patterns, with clear instructions and proper numbering.
- I need short notes aligned to the model paper's topics, with key points, formulas/steps, and brief examples.
- I need the interface to be simple and easy to navigate

System Requirements

- Ability to process PDF lecture materials and past exam papers.
- Integration of NLP libraries for keyword extraction, clustering, and summarization.
- Storage of processed data and generated outputs.
- Web or desktop interface for interaction with the system.

5. Budget

Table 2 - Budget

Type	Cost
Internet use and web hosting	LKR.10,000.00
Training Cost	LKR.10,000.00
Publication Cost	LKR.50,000.00
Stationery	LKR.1,000.00
Additional	LKR.10,000.00
Total	LKR.81,000.00

The cost can vary depending on several factors when implementing the final product.

6. Commercialization

The proposed system has significant commercialization potential as a data-driven academic support platform for higher education. By automating the analysis of past exam papers and generating curriculum-aligned model exam papers and short notes, the system provides students with personalized and exam-focused revision resources. Unlike generic study materials, this solution ensures that learners prepare effectively by concentrating on frequently tested topics and authentic exam structures.

From a commercialization perspective, the platform can be deployed as a subscription-based web or mobile application accessible to students across different universities. A freemium model can be adopted, where basic features such as exam paper analysis and syllabus summaries are provided free of charge, while advanced features including model paper generation, short notes are offered under a premium plan. This approach ensures accessibility to all students while creating a sustainable revenue stream.

In addition to direct student subscriptions, the system can be commercialized through institutional licensing agreements with universities. By integrating the platform into Learning Management Systems (LMS), institutions can provide students with reliable academic support tools, while also gaining valuable insights into curriculum assessment alignment. This dual pathway targeting both individual learners and educational institutions positions the system as a scalable product with wide applicability across disciplines.

Ultimately, the commercialization of this system addresses the growing demand for evidence-based, personalized, and efficient academic support. By bridging the gap between curriculum delivery and examination requirements, it not only enhances student achievement but also strengthens the quality of teaching and assessment at an institutional level.

7. Communication Management Plan

7.1 Communication mediums

- Email: Formal communication, record-keeping.
- Phone Calls: Immediate feedback, personal interaction.
- WhatsApp Messaging: Quick updates, informal communication
- Meeting: Group discussion, decision-making

7.2 Communication Objectives

1. Inform: Provide updates or share information.
2. Persuade: Influence opinions or decisions.
3. Engage: Foster interaction and collaboration.
4. Request: Ask for information or action.
5. Clarify: Resolve misunderstandings or provide explanations.

7.3 Meetings

Table 3 -Meeting

Meeting Type	Attendees	Purpose	Frequency	Agenda Items
Planning Kick-off Meeting	Supervisor, Co-supervisor, All Team Members	Launch project planning, define scope, governance, roles, and identify risks	Once at Project Level	Planning timeline, project scope, charter overview, overall schedule, approach, assumptions, risks, and recap of key decisions and actions.
Executing Kick-off Meeting	Supervisor, Co-supervisor, All Team Members	Launch project execution, ensure clarity on scope, governance, and roles.	Once at Project Level or for major phases	Present Work Plan, Communications Plan, dispute resolution, quality procedures, and recap of key points and risks.
Internal Project Status Meeting	All Team Members	Review project status, assess progress, address risks/issues, and review modifications	Once a week	Project progress, accomplishments, work vs plan, milestones, and deliverables status.
Actual Project Status Meeting	Supervisor, Co-supervisor, All Team Members	Discuss current status, risks, issues, and change requests.	Twice a week	Review change requests, progress, next deliverables, risks, and issues.
Project Review Meeting	Supervisor, Co-supervisor, All Team Members	Review project status, scope changes, re-baselining, and alignment with goals.	Quarterly (Before Proposal, PP1, PP2, Final)	Review milestones, testing progress, risks, panel comments, and other key topics.

Final Project Review Meeting	Supervisor, Co-supervisor, All Team Members	Review project achievements, performance, challenges, and lessons learned.	End of project	Evaluate project outcomes, documentation review
Project Steering Committee (PSC) Meeting	All Team Members	Discuss project status, obtain permissions, and confirm commitments.	Monthly or as needed	Project results, issues, management attention, objectives, budget, and deadlines

8. Risk Management Plan

Risk management is an important part of project planning and execution. The major goal of this risk management strategy is to identify possible hazards to the project's success and to define measures for minimizing them. The strategy addresses risks connected with the project team as well as contacts with the panel and supervisor(s).

Risk Category	**Risk Description**	**Mitigation Strategies**
Team-Related Risks		
Skill Gaps and Knowledge Deficiencies	Lack of necessary skills or knowledge within the team.	Early identification of skill gaps, training sessions, workshops, external experts, mentors.
Team Member Availability	Unexpected unavailability of team members due to illness, emergencies, or commitments.	Cross-training, contingency plans, task reallocation.
Communication Breakdown	Miscommunication or lack of communication among team members.	Clear communication channels, team meetings, project management tools, documentation.
Conflict Among Team Members	Disagreements or conflicts within the team.	Conflict resolution mechanisms, feedback sessions, mediation, open communication, mutual respect.
Project-Related Risks		
Project Scope Creep	Project expansion beyond its original scope.	Adherence to project charter, formal change management process.
Panel/Supervisor-Related Risks		
Misalignment of Expectations	Mismatch between project team's output and panel/supervisor's expectations.	Regular meetings, feedback, alignment with expectations.
Delayed Feedback or Decision Making	Slow or delayed feedback from the panel or supervisor.	Clear deadlines, reminders, proactive feedback seeking.
Changes in Panel/Supervisor's	Unavailability of panel or supervisor due to other commitments.	Advance scheduling, alternative contacts, contingency plans.

Availability		
Scope or Requirement Changes from Panel/Supervisor	Changes to project requirements or scope introduced by the panel or supervisor.	Formal change control process, assessment of impacts, negotiation.

Table 4 - Risk Management Plan

9. Testing

Testing is a vital stage in ensuring that the developed system performs as expected and meets both functional and non-functional requirements. For this project, testing is conducted in multiple phases to evaluate the correctness, reliability, and usability of the proposed framework.

At the initial stage, unit testing is performed on individual agents such as the Ingestor Agent (exam paper preprocessing), Exam Analyst Agent (exam structure modeling), Trend Miner Agent (topic frequency extraction), Generator Agent (model exam paper creation), Mapper Agent (semantic mapping to lecture materials), Notes Agent (short notes generation), and Evaluator Agent (quality validation). This ensures that each agent works independently without errors.

Afterward, integration testing is carried out to verify that these agents interact correctly, particularly the link between past paper analysis, exam structure modeling, semantic mapping, and short note generation. Integration testing also checks smooth data flow across the shared corpus and vector database to ensure consistent outputs.

Once integration is successful, system testing is conducted to evaluate the system as a whole. This includes verifying that the framework satisfies the functional requirements (e.g., accurate exam trend analysis, structured model exam paper generation, curriculum-aligned short notes) as well as non-functional requirements such as performance, reliability, and scalability across different input datasets.

Finally, user acceptance testing (UAT) is performed with sample users to ensure that the generated model exam papers and short notes meet academic expectations, usability standards, and practical revision needs. Feedback gathered from UAT provides valuable insights for refining the outputs before deployment.

Overall, the testing phase ensures that the system is accurate, reliable, and user-friendly, capable of supporting effective exam preparation and aligning academic content with historical exam patterns.

10. Conclusion

In conclusion, this research presents an intelligent academic support framework that addresses the persistent gap between lecture delivery and examination requirements. By systematically analyzing historical exam papers, the system uncovers structural and topical patterns, enabling the generation of realistic model exam papers that mirror authentic exam blueprints. In parallel, the automated production of concise, curriculum-aligned short notes provides students with targeted revision resources focused on the most frequently tested areas.

This dual approach not only enhances the efficiency of student exam preparation but also ensures that revision remains evidence-based and syllabus-oriented. The system stands out by combining long-term trend analysis, exam-structure fidelity, and semantic mapping of lecture content, offering a practical and data-driven alternative to traditional study methods.

The framework's modular Agentic AI design ensures scalability, adaptability, and reliability, making it extendable to other academic modules beyond the DMS course. Ultimately, the proposed framework contributes to improved academic performance by equipping learners with reliable, high-impact resources while also supporting institutions in aligning curriculum and assessment practices. By bridging the gap between taught content and examined content, this research demonstrates the potential of AI-powered educational tools to positively transform the student learning experience.

11.References

[1] Gibbs, G., & Simpson, C. (2004). *Conditions under which assessment supports students' learning*. Learning and Teaching in Higher Education, (1), 3–31.
[2] Nicol, D. (2007). *Assessment for learner self-regulation: Enhancing achievement in the first year using learning technologies*. Assessment & Evaluation in Higher Education, 32(2), 199–207. https://doi.org/10.1080/02602930600801901
[3] Romero, C., & Ventura, S. (2020). *Educational data mining and learning analytics: An updated survey*. Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery, 10(3), e1355. https://doi.org/10.1002/widm.1355
[4] Campos, R., Mangaravite, V., Pasquali, A., & Jatowt, A. (2020). *YAKE! Keyword extraction from single documents using multiple local features*. Information Sciences, 509, 257–289. https://doi.org/10.1016/j.ins.2019.09.013
[5] Reimers, N., & Gurevych, I. (2019). *Sentence-BERT: Sentence embeddings using Siamese BERT-networks*. Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing. https://arxiv.org/abs/1908.10084
[6] Heilman, M., & Smith, N. A. (2010). *Good question! Statistical ranking for question generation*. Human Language Technologies: The 2010 Annual Conference of the NAACL. https://aclanthology.org/N10-1086
[7] Kurdi, A., Leo, J., Parsia, B., Sattler, U., & Al-Emari, S. (2020). *A systematic review of automatic question generation for educational purposes*. International Journal of Artificial Intelligence in Education, 30(1), 121–204. https://doi.org/10.1007/s40593-019-00186-y
[8] Burrows, S., Gurevych, I., & Stein, B. (2015). *The eras and trends of automatic short answer grading*. International Journal of Artificial Intelligence in Education, 25(1), 60–117. https://doi.org/10.1007/s40593-014-0026-8
[9] Sajja, R., Reddy, K., & Prasad, K. (2024). *AI-enabled intelligent assistant for personalized and adaptive learning in higher education*. Information, 15(10), 596. https://doi.org/10.3390/info15100596
[10] Mihalcea, R., & Tarau, P. (2004). *TextRank: Bringing order into text*. Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing. https://aclanthology.org/W04-3252
[11] Lewis, M., Liu, Y., Goyal, N., et al. (2020). *BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension*. Proceedings of ACL 2020. https://arxiv.org/abs/1910.13461
[12] Raffel, C., Shazeer, N., Roberts, A., et al. (2020). *Exploring the limits of transfer learning with a unified text-to-text transformer (T5)*. Journal of Machine Learning Research, 21(140), 1–67. https://arxiv.org/abs/1910.10683
[13] Reddy, S., Gupta, V., & Singh, P. (2024). *Large Language Models for educational content generation and evaluation: Opportunities and challenges*. arXiv preprint. https://arxiv.org/abs/2410.09063
[14] Kurdi, H., El-Sayed, R., & El-Khatib, A. (2021). *Automatic question generation and answer assessment: A survey in technology-enhanced learning*. Smart Learning Environments, 8(1), 1–22. https://doi.org/10.1186/s41039-021-00151-1
[15] Xu, W., Mahmud, R., & Hoo, W. L. (2024). *A systematic literature review: Are automated essay scoring systems competent in real-life education scenarios?* IEEE Access, 12, 77639–77657. https://doi.org/10.1109/ACCESS.2024.3384138
[16] Duan, Z., Li, C., Sun, X., & Ji, H. (2021). *Controllable question generation via answer-aware prompts and difficulty calibration*. arXiv preprint. https://arxiv.org/abs/2106.15313
[17] Backenköhler, M., Mazan, P., & Vreede, G. J. (2018). *A data-driven approach towards a personalized curriculum*. Proceedings of the 11th International Conference on Educational Data Mining (EDM 2018). https://files.eric.ed.gov/fulltext/ED593214.pdf